

Lecture 7 - Team Diversity

What is Team Diversity? - **Team Diversity**. The significant uniqueness of each individual on a team. **Embrace**: Hold closely, accept with respect. Creating inclusive cultures where people feel respected and valued for who they are.

Main Diversity Areas - Religious diversity , Gender diversity , Age diversity , National culture , Race Disabilities

Inclusion - Inclusion is creating an environment where everyone feels respected, valued, and involved.

Belonging - Belonging is the emotional feeling of being accepted, safe, and valued in a group. Key Ideas - Feeling included , Feeling respected , Feeling psychologically safe Importance of Diversity - Improves teamwork , Creates innovation , Brings new ideas , Improves decision making , Encourages creativity , Improves communication , Increases productivity

Religious Diversity - Religious diversity means having people from different religions in a team or organisation.

Benefits of Religious Diversity - Different viewpoints , Better decision making , Better understanding of society , Respect for cultures

Challenges of Religious Diversity - Misunderstandings, Conflicts, Cultural disagreements

How to Manage Religious Diversity - Train employees about policies , Be fair in religious observations , Encourage acceptance , Respect differences , Avoid overreacting to small issues

Do not tolerate serious discrimination

GENDER DIVERSITY - Gender diversity means including people of different genders in a workplace or team.

Benefits of Gender Diversity - Wider talent pool (More skilled employees) , Multiple perspectives (Better ideas) , Enhanced collaboration (Better teamwork) , Better staff retention (Employees stay longer) , Better customer understanding (Understand different customers) , Better reputation (Positive organisation image)

Gender Communication Issues - Gender differences may affect: Communication styles , Perceptions , Opportunities , Team interactions , Sexual harassment issues

AGE DIVERSITY - Age diversity means having employees from different age groups.

Benefits of Age Diversity - Different mindsets , Better flexibility , Better decision making , Knowledge sharing , Fast execution , Work-life balance

Importance of Older Workers - Qualities of Older Workers - Dedicated , Punctual , Honest , Detail-oriented , Good listeners , Pride in work , Organisational skills , Efficient , Mature , Good communication skills

Importance of Young Workers - Benefits of Young Employees - 1. New Energy & Perspective (Fresh ideas , Creative thinking) , 2. Workforce Development (Learn quickly , Easy to train) , 3. Affordability (Lower wages) , 4. Technology Skills (Fast technology adoption, Digital knowledge) 5. Adaptability & Agility (Respond quickly to change)

CULTURE - Definition - Culture refers to: values, beliefs, standards, and behaviors shared by a group of people.

National Culture - The set of values or beliefs that a society considers important and the norms of behavior approved in that society. Changes over time and affects members' values and beliefs

Popular Dimensions of Culture - Language , Time orientation , Use of space , Religion

LANGUAGE - Language is one of the most visible aspects of culture.

Whorfian Hypothesis - Language strongly affects thinking patterns.

Low-Context Culture - Communication is direct and clear. Examples - USA (Northern states) , Canada , Germany , Scandinavia **Characteristics** - Words carry full meaning , Direct communication

High-Context Culture - Communication depends on: body language, context, symbols, relationships. Examples - Sri Lanka , China , Japan , India , Arab countries **Characteristics** - Indirect communication, Hidden meanings, Cultural understanding important

TIME ORIENTATION - Polychronic Culture - Culture with flexible view of time.

Characteristics - Focus on present , Multiple tasks together , Less pressure

Monochronic Culture - Culture with strict time management. **Characteristics** - Linear view of time , Long-term planning , High focus on schedules

USE OF SPACE (**PROXEMICS**) - Proxemics is the study of how people use personal space to communicate. Importance - Personal space differs across cultures , Can affect communication , Can create misunderstandings

RELIGION AS CULTURE - Importance of Religion Religion: affects behavior, ethics, moral values, and economic decisions.

Race - Group of people, often of a common geographic origin, that share genetically transmitted physical characteristics

Racism: The belief that inherited characteristics affect individual's behavior or abilities.

Disabilities - Challenges : Communication difficulties , Emotional misunderstandings , Interaction barrier Importance - Teams should support equal participation for disabled individuals.

BENEFITS OF DIVERSITY - Talents & skills(Different abilities) , Innovation (New creative ideas) , Language skills (Better communication) , Larger talent pool (More employees available) , Better performance(Improved productivity)

CHALLENGES OF DIVERSITY - Discrimination(Unfair treatment) , Stereotyping (Judging groups unfairly) , High turnover(Employees leave) , Absenteeism (Frequent absence) , Conflicts (Team disagreements) , Tension (Stress in workplace) , Productivity decrease (Reduced performance)

DISCRIMINATION - Discrimination means treating someone unfairly because of their characteristics. - Age , Gender , Race , Religion ,Disability , Pregnancy ,Sexual orientation

STEREOTYPING - Oversimplified, usually negative attitude people hold toward those outside their own experience who are different.

Causes of Stereotyping - Prejudice , Social perception , Expectations , Lack of knowledge.

Effects of Stereotyping - Unfair judgments , Conflicts , Discrimination , Poor teamwork

HOW TO OVERCOME DIVERSITY ISSUES - 1. Planning & Implementation - Managers should: create diversity plans, encourage appreciation, support inclusion. / 2. Conflict Resolution Skills - Managers must: solve conflicts quickly, reduce misunderstandings, maintain productivity. 3. Communication - Good communication: encourages openness, builds trust, solves problems, improves teamwork.

CULTURE SHOCK - Culture shock is confusion or discomfort experienced in a different culture. CROSS-CULTURAL COMMUNICATION - Communication between people from different cultures.

What is **Inclusion**? - Creating environments where everyone: feels respected, has equal opportunities, and can contribute successfully.

What is **Belonging**? - Feeling: accepted, safe, valued, inside a team.

Benefits of Inclusion & Belonging - Better teamwork , Better communication , Psychological safety , Strong trust , Less conflict , Better collaboration

Belbin Team Roles - Inclusive teams allow different roles and personalities to work successfully.

Profession - A calling requiring specialized knowledge and intensive academic preparation ex ; Doctor , Engineer , Lawyer

Professional - Person engaged in a learned profession, conforming to technical/ethical standards

Professionalism - Conduct, aims, or qualities that characterize a professional

WHO IS A PROFESSIONAL? - Main Characteristics -1. Expert Knowledge -Professionals must have: technical knowledge, education, and skills. 2. Autonomy - Definition -Ability to work independently and make professional decisions. 3. Moral Responsibilities - Definition - Professionals have ethical duties beyond ordinary people. - Examples -Honesty , Responsibility , Public safety , Fairness

Amateur vs Professional - **Amateur** - Works for interest, Informal skills , No strict standards , No professional ethics - **Professional** - Works as career , Specialized skills , Follows standards, Uses ethics and codes

ATTRIBUTES OF A PROFESSION - 1. Education Profession requires: academic preparation, training, certifications. 2. Accreditation - Definition - Official recognition of education programs or qualifications. 3. Skills Development - Professionals continuously improve skills. 4. Licensing - Official permission to practice profession. 5. Certification - Proof of competence and professional qualifications. 6. Code of Ethics - Professionals must follow ethical rules and standards. 7. Professional Development - Continuous learning and improvement. 8. Professional Associations - Organizations that support professions and public

interests.

Ethics - Definition: Set of beliefs about right or wrong behavior Ethical behavior: Conforms to generally accepted social norms (universally accepted) - Example: Lying and cheating are unethical - Why Ethics? Helps make moral decisions , Assumes people are rational & free to choose , Describes how people act and should act

CODE OF ETHICS - A code of ethics is a set of rules and standards that guide professional behavior. **Professional Ethics** - Professional ethics include: personal standards, organizational standards, corporate standards.

CSLL CODE OF ETHICS - 1. **Professionalism** Professionals should: improve society, respect members, protect safety and welfare. 2. **Professional Development** Professionals should: improve careers, help colleagues develop. 3. **Added Value to Sri Lanka** 4. **Honesty** -avoid misrepresentation 5. **Competence** - Professionals should:work carefully, perform only skilled tasks, maintain quality standards.

IEEE CODE OF ETHICS - Professionals should: Accept responsibility , Protect safety and welfare , Avoid conflicts of interest , Be honest , Reject bribery

Professional Associations - Organizations that support professions and public interests.

Organization - CSSL(Computer Society of Sri Lanka), IET (Institute of Engineering and Technology) , IEEE (Institute of Electrical and Electronics Engineers) , ACM (Association for Computing Machinery), BCS(British Computer Society), IESL (Institution of Engineers Sri Lanka)

RESPONSIBILITIES OF COMPUTING PROFESSIONALS - 1. Responsibility to Clients & Users Professionals must: provide quality services, protect users, ensure safety. 2. Responsibility to Employers - Professionals should: work honestly, protect company interests, maintain confidentiality. 3. Responsibility to Other Professionals - Professionals must: respect colleagues, cooperate fairly, support teamwork. 4. Responsibility to the Public - Professionals must: protect society, avoid harm, ensure public safety.

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Lecture 6

What is Leadership? - Leadership is: Knowing yourself, Having a clear vision , Communicating that vision Building trust Taking effective actions.

Warren Bennis Definition - Leadership cannot exist without the cooperation and inclusion of employees.

Why Leadership is Controversial

Stable Environment - Captain of a ship , Predictable , Controlled

Volatile Environment - Rafting on white water , Unpredictable , Fast-changing

Different situations need different leadership styles.

Leadership Theories

Great Man Theory - Leaders are exceptional people, born with innate qualities, destined to lead. This concept was primarily male, military and western(Exceptional people are destined to lead.)

Trait Theory - Research on traits or qualities associated with leadership are numerous • But, Traits are hard to measure. Ex: honesty, integrity

Leadership Traits - Adaptable , Ambitious , Assertive , Cooperative , Decisive , Dependable , Dominant , Energetic , Persistent , Self-confident , Stress tolerant , Responsible

Leadership Skills - Intelligent , Creative , Diplomatic , Fluent speaker , Organised , Persuasive , Socially skilled , Knowledgeable

Situational Leadership Theory (SLT) - Developed by: Hersey & Blanchard Main Idea - Leadership style changes based on: Follower ability , Follower willingness

Leadership Styles - Unable + Unwilling (Directive (Telling)) , Unable + Willing (Coaching (Selling)) , Able + Unwilling (Supportive (Participating)) , Able + Willing (Delegating (Observing))

Transformational Leadership Theory - Developed by: Bass & Avolio (1994) Main Idea - Leaders: Inspire followers , Build trust , Encourage creativity , Support personal growth - Focus : People work for: Team goals , Organisation goals , Society goals (Not only personal rewards.)

Team Decision-Making -

Decision: Making up your mind about something, or a position or opinion after consideration.

Decision Making: Cognitive process of choosing btw alternatives using memory, thinking, & evaluation.

Team Decision Making: Process where a team selects alternatives important for team performance.

Why Learn Decision Making? - Make informed choices , Involve everyone , Justify decisions , Improve teamwork

Decision Making Process -Step 1 - **Create Constructive Environment** Focus on: Goals , Right people , Opinions - Step 2 - **Generate Alternatives** - Use: Ideas , Brainstorming , Different perspectives - Step 3 - **Explore Alternatives** - Check: Risks , Impacts , Validation - Step 4 - **Choose Best Alternative** - Use: Grid Analysis , Decision Trees - Step 5 - **Check Decision** - Review assumptions, Use experience , Avoid bias - Step 6 - **Communicate & Take Action** - Explain decision, Implement it

Three Areas Supporting Decision Making - Environment , Methods , Tools

ENVIRONMENTS FOR DECISION MAKING - Features of a Thinking Environment (10 key features): 1.**Attention** - Listen with respect & interest to team members 2.**Incisive Questions** - Remove assumptions that limit ideas 3.**Equality** - Treat each member as a thinking peer

4.**Appreciation** - Practice 5:1 ratio of appreciation to criticism 5.**Ease** - Ease creates, urgency destroys 6.**Encouragement** - Collaborate, don't compete 7.**Feelings** - Allow emotional release to restore thinking 8.**Information** - Provide accurate picture of reality 9.**Place** - Create physical environment that says "You matter" 10.**Diversity** - Add quality through differences

METHODS FOR DECISION MAKING - What Makes a Best Decision? - Wouldn't have been thought of by an individual alone , A sound solution to the problem , Based on unbiased input from each team member , Addresses the team's goal

7 Methods OF Decision Making: **Decision by Consensus**. When: Enough time available, team is skilled, high commitment needed **Decision by Majority Vote**. When: Time constraints exist, moderately high commitment needed **Decision by Minority Vote (EXCO)**. When: Limited time, clear **minority group choice**, moderately low commitment needed **Decision by Authority After Group Discussion**. When: Time allows interact but not full agreement, moderately low commitment needed **Decision by Averaging Individuals' Opinions**. When: Limitedtime, team participant required but not lengthy interact, low commitment needed **Decision by Expert**. When: Highly dependent on specific expertise, clear expert choice **Decision by Authority Without Group Discussion**. When: Simple, routine decisions with little time available

Disadvantages of Team Decision Making - 1. **Time Consumption** - Slower than individual decision making 2.**Less Recognition** - Hard to identify individual contributions in team effort

3.**Conformity** - People conform to consensus, suppressing different opinions

Advantages of Team Decision Making - 1.**Greater Knowledge** - More people = more ideas, experience, and perspectives 2.**Shared Responsibility** - People more willing to take risks together

3.**Higher Commitment** - Contributors feel committed to implementing the solution

TOOLS FOR DECISION MAKING - Why Use Tools? Tools assist teams in formulating and reaching decisions effectively.

Key Tools: 1. **Brainstorming** - Group CREATIVITY TECHNIQUE to generate ideas spontaneously for a specific problem 2. **Affinity Grouping** - Grouping unstructured brainstorming ideas based on their natural relationships 3. **Multi-voting** - Used when there are too many options - **Process**: Each member gets multiple votes (1/3 or 1/4 of total used), run multiple rounds until top options remain - Example: Sri Lankan Presidential Elections use preference voting system

4.**Criteria Matrix (Decision Matrix)** - List of values in rows and columns to systematically analyze and rate options based on decision criteria 5. **Six Thinking Hats** Looking at a decision from different perspectives

White Hat: Data, facts, information known or needed

Red Hat: Feelings, hunches, instinct and intuition

Black Hat: Difficulties, potential problems, why something may not work

Yellow Hat: Values and benefits, why something may work

Blue Hat: Manage process, next steps, action plans

Green Hat: Creativity, solutions, alternatives, new ideas

CHARACTERISTICS OF ORGANIZATIONAL CULTURE - 1. **Observed Behavioral Regularities** - Meaning - Common behaviors and language. - Examples - Greetings , Technical terms , Work habits 2. **Norms** - Definition - Expected standards of behavior. Examples - Work targets - Attendance expectations 3.**Dominant values** - Definition - Main beliefs important to organization. Examples - Quality , Efficiency , Low absenteeism 4. **Philosophy** - Definition - Organization beliefs about employees and customers 5. **Rules** - Definition - Formal guidelines employees follow. 6. **Organizational Climate** - Definition - Overall atmosphere of workplace. Includes Interaction style - Employee behavior , Customer relationships

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Name : -

IT Number :-

Lecture 8

Power: The probability that one actor within a social relationship will be in a position to carry out his own will despite resistance. ability to influence another

Authority - right to influence another person

Difference Between Power & Authority - **Power** - Ability to influence , Can exist without position , Informal or formal -

Authority - Right to influence , Comes from official position , Usually formal

FIVE FORMS OF POWER - **Reward Power** - The ability to provide rewards or incentives to influence behavior Workplace

Examples: Giving promotions, raises, or bonuses , Providing recognition and praise , Offering desirable assignments or opportunities Granting additional benefits or perks Advantages

Motivates employees, Improves performance Disadvantages - Employees may only work for rewards

Legitimate Power Power that is granted by virtue of one's position or title (power that comes with the job title) Workplace

Examples: Authority that comes from official job title or rank , Power derived from organizational hierarchy , Legal or formal authority to make decisions.Rights & responsibilities assigned by the organization Advantages -Creates order ,Employees follow instructions Disadvantages -Depends heavily on position

Coercive Power - Power that comes from a person's ability to punish Someone else if expectations are not met (To achieve compliance). Ex: Ability to fire, demote, or suspend employees

Withholding resources or opportunities ,Imposing penalties or negative consequences Advantages - Controls discipline

Disadvantages - Creates fear ,Reduces motivation

Expert Power - This comes from having specialized knowledge, skills, or expertise that others need or value. Workplace

Examples: Technical expertise or specialized skills Knowledge that others depend on Professional qualifications or certifications

Experience and competency in specific areas - Advantages - Gains respect, Creates trust - Disadvantages - Depends on expertise level

Referent (Charismatic) Power - This is thought of as charisma, charm, or admiration. Referent power comes from one person liking and respecting another, and strongly identifying with that person in some way (personal connection). Workplace Examples: Personal charisma and likability , Respect earned through character and integrity , Influence based on personal relationships , Power derived from being a role model.

CONSEQUENCES OF POWER - Expert Power (Commitment) , Referent Power (Commitment) , Legitimate Power (Compliance), Reward Power(Compliance) , Coercive Power (Resistance)

Explanation of Consequences -Commitment - Employees willingly support leader. **Compliance** - Employees obey rules without strong emotional support. **Resistance** - Employees oppose or dislike authority.

POLITICAL BEHAVIOR IN ORGANIZATIONS - Political behavior refers to actions not officially authorized by the organization but used to influence others to achieve personal goals. **Characteristics of Organizational Politics** - Informal behavior, Influence others , Achieve personal goals , Common in organizations

Importance of Political Behavior -Helps gain influence , Helps achieve goals , Improves networking , Helps career growth

Dale Carnegie Concept- Main Idea - Make other people feel important sincerely. Key Words - Please , Thank you , Respect ,Appreciation

POLITICAL STRATEGIES FOR ATTAINING POWER - 1. **Taking Counsel** - Meaning - Seeking advice from others. 2. **Maintaining Maneuverability** - Meaning - Keeping flexibility in decisions. 3. **Promoting Limited Communication** - Meaning - Sharing only necessary information. 4. **Exhibiting Confidence** - Meaning - Showing confidence to gain trust. 5. **Controlling Access to Information & Persons** - Meaning - Managing who receives information. 6. **Making Activities Central & Non-substitutable** - Meaning - Becoming essential to organization operations. 7. **Creating Sponsor-Protégé Relationship** - Meaning - Senior person supports junior employee. 8. **Stimulating Competition** - Meaning - Encouraging competition among workers. 9. **Neutralizing Potential Opposition** - Meaning -Reducing threats from opponents. Example - Using media or public opinion against critics. 10. **Making Strategic Replacements** - Meaning - Replacing people strategically to gain advantage 11. **Committing the Uncommitted** - Meaning - Making disengaged employees become committed 12. **Forming a Winning Coalition** - Definition - Building alliances with groups to achieve goals. Example - Maithripala Sirisena's 2015 election coalition. 13. **Developing Expertise** - Meaning - Increasing knowledge and skills .14. **Building Personal Stature** - Meaning - Improving personal reputation and image. 15. **Employing Trade-Offs** - Meaning - Making compromises for benefits. 16. **Using Research Data** - Meaning -Using evidence to support arguments. 17. **Restricting Communication About Intentions** - Meaning - Keeping real intentions private. 18. **Withdrawing from Petty Disputes** - Meaning - Avoiding unnecessary conflicts.

ORGANIZATIONAL CULTURE - Organizational culture is: shared values, beliefs, assumptions, and behaviors developed by a group over time. Importance of Organizational Culture - Guides employee behavior , Creates teamwork , Improves communication , Builds organizational identity , Increases performance

ACM CODE OF ETHICS - Contribute to society , Avoid harm , Be honest . Be trustworthy . Avoid discrimination , Respect privacy , Respect confidentiality , Respect intellectual property

ACM SOFTWARE ENGINEERING CODE - PUBLIC , CLIENT & EMPLOYER , PRODUCT , JUDGMENT , MANAGEMENT , PROFESSION , COLLEAGUES , SELF

PURPOSE OF PROFESSIONAL CODES – Inspire – Encourage ethical behavior , Educate – Teach correct conduct , Guide - Help make decisions , Responsibility - Define duties , Discipline - Control misconduct

Lecture 10

ETHICAL HIERARCHY – **Humanity** (Integrity, fairness, care) , **Professionalism** (Standards and ethics) , **Profession** (Rules and duties of profession)

What are **Ethics**? - Ethics are guidelines that help decide: right behavior, wrong behavior, moral decisions.

KEY IDEAS OF IT ETHICS - Technology creates new moral issues , Privacy can be lost easily , Ethical decisions need long-term thinking , Users must be treated fairly , Inclusion reduces discrimination , Ethical culture encourages responsibility

PROFESSIONAL RESPONSIBILITIES - 1. Responsibility to Clients - Professionals should: provide reliable systems, communicate honestly, protect client data. 2. Responsibility to Employers - Professionals should: follow security policies, report threats, avoid misuse of company resources. 3. Responsibility to Colleagues - Professionals should: respect others, avoid discrimination, support teamwork, maintain healthy workplace culture. 4. Responsibility to Society - Professionals should: avoid harmful systems, respect human rights, support sustainability, reduce bias and discrimination.

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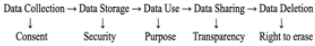
ETHICAL DECISION-MAKING MODEL – step 1 - Identify ethical problem , Step 2 - Identify stakeholders , Step 3 - Apply ethical theories , Step 4 - Evaluate consequences , Step 5 - Make decision

STAKEHOLDER ANALYSIS - Stakeholder analysis means identifying everyone affected by a decision.

Possible Stakeholders – Users , Employees , Customers , Governments , Society , Vulnerable groups

ETHICAL THEORIES - Utilitarianism (Outcome-Based) - Choose the action that benefits the greatest number of people. – Advantages - Focuses on social benefit , Useful for large-scale decisions. - Disadvantages - Minority groups may suffer unfairly , Deontology (Duty-Based) - Some actions are always right or wrong regardless of outcome. – Advantages - Protects human rights , Encourages fairness - Disadvantages - Sometimes ignores overall benefit

GDPR (General Data Protection Regulation) - Strongest data privacy law created by the European Union (EU) in 2018. Purpose of GDPR - Protect: collection, storage, use, sharing of personal data.



USER RIGHTS UNDER GDPR - 1. Right to Access , 2. Right to Correction , 3. Right to Deletion (Right to be Forgotten) , 4. Right to Data Portability

CCPA (California Consumer Privacy Act) - Main Features - Users can stop data selling , Companies must explain collected data

Personal Data Protection Act (PDPA) No. 9 of 2022 - Consent required , Users can correct/delete data , Organizations must secure data , Data Protection Authority enforces law

LEGAL RESPONSIBILITIES OF IT PROFESSIONALS - Professionals should: protect personal data, follow laws, secure systems, use encryption, maintain audit records.

PRIVACY PRINCIPLES - Data Minimization(Collect only needed data), Purpose Limitation(Use data only for stated reason) , Data Retention (Delete unnecessary data) , Transparency(Keep users informed)

Lecture 11

What is **Sustainability**? - Meeting the needs of the present without compromising the ability of future generations to meet their own needs



WHY ETHICS IS IMPORTANT IN COMPUTING - Computing systems can: track people, manipulate users, collect private data, influence society.

ETHICAL DILEMMAS IN IT - Situations where difficult moral decisions must be made.

IMPORTANT ETHICAL IDEAS - Developers must predict consequences , Technology should not create inequality , Fairness and inclusion are important , Psychological safety should exist in workplaces

BIASED SYSTEMS - Systems that unfairly favor or disadvantage certain groups. – Cause - Biased cultures or biased training data.

SUSTAINABILITY IN IT - Using technology in ways that protect: environment, society, and future generations.

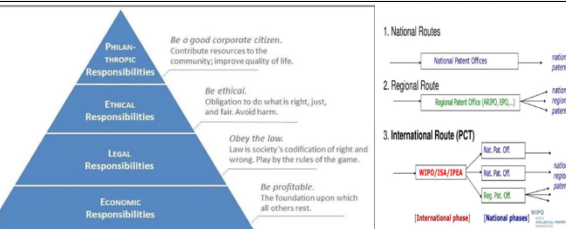
GLOBAL SUSTAINABILITY ISSUES - 1. **E-Waste** - Discarded electronic devices and hardware. Causes - Fast upgrades , Short device lifespan , Hardware obsolescence 2. **Hazardous Materials** - Electronics manufacturing uses dangerous chemicals and materials. 3. **Data Centers & Energy Usage** - Large data centers consume huge amounts of electricity. Result - High carbon emissions. 4. **Cryptocurrency Mining** - Uses massive computing power. Result - Very high environmental impact. 5. **Fast Upgrade Culture** - People replace phones/devices every 1–2 years. Results - More electronic waste.

SOLUTIONS FOR SUSTAINABILITY - 1. **Green Computing** - Designing energy-efficient software and hardware. 2. **Circular Economy** - Reusing, repairing, refurbishing, and recycling products. 3. **UN Sustainable Development Goals (SDGs)** - Important goals include: Climate action , Responsible consumption , Sustainable development 4. **Repair Culture** - Encouraging repair instead of frequent replacement.

IT LAWS & DATA PROTECTION - GDPR (General Data Protection Regulation) - Key Ideas - Companies must be transparent , Users have strong rights , Violations lead to heavy fines

SRI LANKA PDPA - Personal Data Protection Act - Key Points - Consent needed before collecting data , Users can correct/delete data , Organizations must secure data , Violations cause penalties

CORPORATE ACCOUNTABILITY - Organizations must take responsibility for: ethical impact, environmental impact, social impact. Fairness - Systems should treat all users equally. Inclusion - Everyone should feel respected and involved. Belonging - People should feel accepted and safe. - Psychological Safety - Environment where people feel safe to: speak, share ideas, make mistakes, without fear.



Lecture 12

What is **Corporate Responsibility**? – Definition - Corporate responsibility means companies must consider: social impact, ethical behavior, environmental effects, and public welfare while conducting business.

WHAT IS **DIGITAL INCLUSION**? - Digital inclusion means ensuring everyone can: access, use, and benefit from digital technologies equally.

WHY CORPORATE RESPONSIBILITY & DIGITAL INCLUSION MATTER - Technology affects: education, healthcare, banking, transport, communication, government services. Because companies control many digital systems, their decisions affect: who gets opportunities, who gets excluded, and who may be harmed.

SOCIAL RESPONSIBILITIES - Social responsibility mainly includes: Ethical responsibilities , Discretionary responsibilities

WHY DIGITAL INCLUSION MATTERS - Digital inclusion ensures: equal access, equal opportunity, social participation.

IMPORTANT IDEA FOR SRI LANKA - Digital inequality exists between: Urban vs Rural areas , English vs Sinhala/Tamil speakers , Younger vs Older people

DIGITAL DIVIDE - Gap between people who have access to digital technology and those who do not. - CAUSES OF DIGITAL DIVIDE - Low income , Poor internet access , Lack of devices , Language barriers , Disability , Lack of digital skills

COMPONENTS OF DIGITAL INCLUSION - 1. ACCESS – Definition - Availability of: internet, devices, network connectivity. 2. AFFORDABILITY – Definition - Technology and internet services should be financially affordable. 3. ABILITY – Definition - People should have digital literacy and technical skills 4. ACCESSIBILITY - Definition Technology should support people with disabilities. Examples - Screen readers , Voice assistants , Larger text , Keyboard navigation 5. AWARENESS – Definition - Understanding benefits and uses of digital technologies.

CORPORATE RESPONSIBILITIES IN IT - RESPONSIBILITY TOWARD CLIENTS - Companies should: build secure systems, protect user data, provide accessible interfaces, create inclusive systems. RESPONSIBILITY TOWARD EMPLOYERS - Employees should: follow policies, report security risks, avoid misuse of company data. RESPONSIBILITY TOWARD COLLEAGUES - Professionals should: support teamwork, avoid discrimination, encourage belonging, create inclusive workplaces. RESPONSIBILITY TOWARD SOCIETY - Companies should: reduce inequality, avoid biased algorithms, create fair systems, reduce digital divide.

ETHICAL CORPORATE BEHAVIOR - Includes: transparency, honesty, fairness, accountability, respect for users.

FAIRNESS IN DIGITAL SYSTEMS – Definition Systems should not unfairly favor or disadvantage groups - Examples of Unfair Systems : Biased AI hiring systems , Language discrimination , Inaccessible government websites

ACCESSIBILITY : Definition - Designing systems usable by all people including disabled individuals. BENEFITS OF ACCESSIBILITY - Equal opportunity , Better inclusion , Wider user access , Improved user experience

Lecture 13 - Intellectual Property (IP)

Why IP System?

1. Enables people to earn: Recognition - Credit for their creations Financial benefit - Profit from what they invent or create
2. Aims to foster a balanced environment in which creativity & innovation can flourish

What is **Intellectual Property (IP)**?

Definition: Any intangible asset that consists of human knowledge and ideas. Includes: Patents, copyrights, trade secrets, trademarks, contracts

Purpose: To encourage ingenuity and protect creative ventures

Value Challenge: Difficult to objectively value IP assets when internally generated. Can be included in balance sheet if acquired (using acquisition cost).

International Treaties that Sri Lanka is a party to - The Paris Convention for the Protection of Industrial Property (since 1952) , The Madrid Agreement for the Repression of False or Deceptive Indication of Source on goods (since 1952) , The Nairobi Treaty for the Protection of Olympic Symbol (since 1984) , The Patent Co-operation Treaty (since 1982) , The Berne Convention (since 1959) , The Universal Copyright Convention (since 1983) , The Convention establishing the World , Intellectual Property Organization (since 1978) , The TRIPS Agreement (since 1995) , The Trademarks Law Treaty (since 1996)

Act No. 36 aims to, Promote the national creativity , Protect the national creative efforts , Facilitate & rationalize d trade , Enhancement of investment , Protect the consumer interests , Integrate the National economyinto the knowledge driven global scenario

IP System of Sri Lanka - The first patent was granted on Nov 22, 1860 in SL under the British Inventor's Ordinance of 1859 , The present IP system operates according to the Act No 36 of 2003 (effect from Nov 12, 2003) , Covers patents, trademarks, copyrights, industrial designs, unfair competition, etc.

Proprietary licenses

- Most proprietary (\$) software is licensed, NOT sold
- Publisher grants "license" to use a copy of s/w
- Ownership stays with s/w publisher of the copies
- Without accepting license, end user cannot use s/w

Are EULAs enforceable? - never fully challenged in court (without a settlement)

Licensing of an IP - License is a permission that an IP owner grant to another person to use the IP on agreed terms and conditions, while he continues to retain ownership of the IP. Licensing creates an income source It establishes a legal framework for transfer of technology to a wider group of researchers and engineers

Creates market presence for the technology or trademark

Trademarks - Protects: Any word, name, symbol, device, or combination , use to distinguish goods from others Protects Against: Others using the mark, Likelihood of confusion, Dilution , Excluded: Use in other industries or geographic areas Requirements: Use mark in commerce or register with intent to use, Must maintain quality control over goods Term: 10 years renewable (no upper limit)

Copyright - Protects: Expression of ideas in a medium, but NOT the ideas themselves Protects Against: Reproduction, Copy distribution, Derivative work creation (Does NOT protect against independent creation of same/similar work) Requirements: Original work, Fixed in tangible form Term: Author's life + 70 years , Cost: Simple registration process

Reverse Engineering - Definition: Taking something apart to see how it works - In S/W: Usually means decompilation (converting object code to source code)- Legal Issues: Almost always violates copyright (based on copying software) unless right to do so is explicitly granted - Exceptions: US and European laws permit it in some cases for interoperability

Trade Secrets - Protects: Concept, idea, information, or innovation , Protects Against: Misappropriation (NOT independent creation) , Requirements: Info not generally known or available, Reasonable effort to maintain secrecy, Company derives economic value from secrecy , Term: No predefined limit , Cost: No registration or examination needed

How to Protect Your Intellectual Property

1. **Register Appropriate IP Protection** - Strongest protect comes from registrat. Puts claim in public view, discouraging unauthorized use.
2. **Trademark Registration** - Fill trademark application on main brand & Apply for registration 3. **Patent or Provisional Patent** - Fill provisional patent application & apply for full patent to protect create 4. **Copyright Registration** - Register your copyright for stronger legal protection 5. **Protect Trade Secrets** - Establish policies and procedures to demonstrate ownership

Fair Use - Definition: Copyright law does allow limited copying, distribution, and display of copyrighted works without author's permission under certain conditions. Copyright law gives authors and creators certain exclusive rights to control the use of their works.

Four Standards for Fair Use - The purpose and character of the use. Including whether such use is of a commercial nature or is for nonprofit educational purposes , The nature of the copyrighted work , The amount and substantiality of the portion used in relation to the copyrighted work as a whole , The effect of use on the potential market for or value of the work.

Types of Intellectual Property - 1. Patents - Protects inventions and innovations , Gives exclusive rights to make, use, and sell the invention Term: Limited period (typically 20 years) 2. Trademarks - Protects brand names, logos, symbols Distinguishes goods/services from others Example: Champagne (can only be called Champagne if from Champagne region of France) 3. Copyright - Protects original creative works (books, music, software, art) Automatic protection when work is created 4. Trade Secrets - Protects confidential business information - Gives competitive advantage No registration needed 5. Geographical Indications - Protects products from specific geographic locations , Quality and reputation linked to place of origin

How to Apply for a Patent - There are 3 routes to apply for a patent: 1. **National Route** - Apply directly to National Patent Offices in your country Result: National patents (valid only in that country) 2. **Regional Route** - Apply to Regional Patent Offices (like ARIPO, EPO, etc.) Result: National and regional patents (valid in multiple countries in that region) 3. **International Route** (PCT - Patent Cooperation Treaty) - Apply through WIPO/ISA/IEPA (World Intellectual Property Organization). **Two phases** : - **International phase** - Single application processed by WIPO - **National phases** - After international phase, file in specific National or Regional Patent Offices Result: National, regional patents in selected countries Purpose: Simplifies filing patents in multiple countries through one initial application

Why file a patent application? - Gives exclusive control over your product (prevents copying & knockoffs) , Allows licensing or selling patent rights for profit or collaboration , Helps attract investors and funding , Builds reputation and recognition. Motivates academic researchers through recognition.

Who has rights to a patent? Inventor(s): Hold the original rights. , **Employee**: Rights depend on employment agreement. , **Contractor**: Rights depend on contract terms. Applies to private/government employees and university members (staff & students) , IP rights are governed by Intellectual Property (IP) policy

Software Piracy - Software piracy is the unauthorized and illegal , duplication of copyrighted software , Covers in Sri Lanka Intellectual Property Act, No. 36 of 2003 , You do NOT have the right to copy, loan, rent, or distribute software unless otherwise , specified in the license. Statistics: The commercial value of pirated software is \$19 billion in North America & Western Europe and has reached \$27.3 billion in the rest of the world, According to the 2018 Global Software Survey , 37% of software installed on personal computers is unlicensed software.

Patentability Criteria - Three Principal Requirements for a Patent: 1. **Novelty** - New characteristic that was NOT known to the public before the filing date or priority date Must be completely new, not previously disclosed 2. **Utility** (Industrial Application) Invention needs to be able to be used or made in any kind of industry Must have practical application 3. **Non-obviousness** (Inventive Step) Invention should NOT be obvious to a person skilled in the art Person skilled in the art = Technician having all knowledge of the technical field of the invention. Must involve creative/inventive thinking, not just common improvements Summary: To get a patent, your invention must be new, useful, and not obvious to experts in that field.

Patents - What Can Be Patented? New inventions , Innovative processes , Improvements to existing , products/processes

Non-patentable subject matters: Mathematical formulae , Naturally occurring organisms , Laws of nature , Abstract ideas , Natural phenomenon , Patentable subject matter : Product , Process

When to file a patent application? - Follow **first-to-file principle** (whoever files first gets the patent). Avoid public disclosure before filing, as it becomes prior art (loses novelty). **Grace period**: Limited time to file after public disclosure in some cases. SL has 12 month grace period

Where to file the patent application? Filing a national , application with a National office (NIPO) Filing a regional application with a Regional office Filing an international application using the PCT (Patent Cooperation Treaty)

Local filing: Lower cost. Can use local language , Can obtain priority date fast Foreign filing: Consider market, legal requirements, and enforcement abroad Consider customers/users & competitors. Look for possible licensees.

Five major types of software piracy

Soft-lifting: Act of purchasing a single copy of software and downloading onto multiple computers, despite agreement terms stating the software must only be downloaded once

Client-Server Overuse: No. of users of a particular s/w exceeds the no. of licenses. Multiple users accessing s/w simultaneously without proper licenses

Internet Piracy: Occurs when s/w is downloaded from any site other than the publisher's official distribution stream

Hard-Disk Loading: Happens when a business installs unauthorized copies of s/w onto the hard disks of any computers it sells

Software Counterfeiting: Illegal reproduction and distribution of copyrighted material. Fake copies sold as genuine

Open-Source Software Source code freely distributed and accessible to everyone. Key Features: Encourages examining and improving others' code New versions appear more frequently Eliminates tension between copyright law and helping others Anyone can make money by providing support Best support providers are rewarded Ex: Apache web servers, Linux OS